



Figure 4: LED on



Figure 5: LED off

User LEDs on MinnowBoard

There are a total of five LEDs on MinnowBoard as shown in Figure 3. Out of those five, two are user LEDs, namely D11 and D12, but by default, they cannot be used as user LEDs. You need to change their default trigger for that.

The user LEDs are accessible via the user space in Linux at the location given below, on the filesystem:

```
/sys/class/leds
```

There is one directory per user LED, named as shown below:

```
/sys/class/leds/minnow_led0
/sys/class/leds/minnow_led1
```

Here, led0 is D11 while led1 is D12. Inside each one of those directories, there is a file named 'brightness'. If you write a '1' or a '0' to this file, then you can control the status of that LED, i.e., toggle it ON or OFF, respectively.

Changing the default trigger

In a nutshell, what we are trying to do is to access the on board user LEDs via the user space in Linux. To be more precise, we are trying to use the `sysfs` interface. `Sysfs` is

a virtual filesystem, which enumerates the devices and busses attached to the system (board, in our case) into a file system hierarchy that can be accessed from user space. It is generated by the kernel and always mounted at `/sys`. As discussed earlier, the trigger for D11 and D12 LED(s) has to be changed to make them function as user LEDs. I would describe 'trigger' as the API used to link an LED to an 'event' in kernel space. Here, 'event' could be a microSD card or Ethernet activity, heartbeat, power, etc. To understand this better, let's look at LED D11, for example. The trigger for D11 has been set by default to 'heartbeat'. So, to use it as a user LED, we have to change the trigger for it to 'none' as shown below:

```
echo none > /sys/class/leds/minnow_led0/trigger
```

Toggling the user LED

This is the best part. Write the following commands in your terminal to toggle the D11 user LED (the first one is for turning ON and the latter for turning OFF):

```
echo 1 > /sys/class/leds/minnow_led0/brightness
echo 0 > /sys/class/leds/minnow_led0/brightness
```

You should get the output shown in Figures 4 and 5. 

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